The single-spin (Option 2) preconditioner in SBD/gdb: what was wrong and what changed

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Abstract

The projected (single-spin) Davidson solver in `gdb` converged to energies *above* the true root, with the discrepancy growing with the size of the projected space, while the spin of every converged root remained exactly correct. This note documents the cause — the preconditioner used only the determinant-diagonal part of the projected Hamiltonian, silently discarding the intra-configuration exchange elements — and the fix, which builds the exact block diagonal of $V^{\mathsf{T}}HV$. Convergence became monotone and the iteration count at $K = 113\,394$ dropped from a stall to 25 matrix–vector products. Commit `f0ff1b8`.

1 Setting

Option 2 produces wavefunctions of a single total spin S by projecting the (Anderson spin-completed) determinant space onto the target-S CSF space *before* diagonalising. Let

- H be the Hamiltonian in the determinant basis, of dimension N_{det};
- V be the determinant → CSF map, of shape $N_{\mathrm{det}} \times K$, with K the number of target-S CSFs.

V is block diagonal by spatial configuration. Because $\hat{S}^2$ does not mix different spatial configurations, each block can be built independently: for a configuration with n_{open} singly occupied orbitals one enumerates its Sz-preserving spin arrangements, diagonalises the small canonical $\hat{S}^2$ matrix, and keeps the eigenvectors with eigenvalue $S(S+1)$. Blocks are cached by n_{open}, since the canonical $\hat{S}^2$ block does not depend on which orbitals are involved.

The solver then runs a block Davidson–Liu iteration on the projected operator

$$\tilde{H} \;=\; V^{\mathsf{T}}HV, \tag{1}$$

applied matrix-free as $x_{\mathrm{csf}} \to Vx_{\mathrm{csf}} \to H(\cdot) \to V^{\mathsf{T}}(\cdot)$. Every iterate is a combination of target-S CSFs, so spin purity is structural: it holds for any preconditioner, however bad.

2 The defect

Davidson needs a cheap approximation to the diagonal of the operator it is solving, used to precondition the residual:

$$t_i \;=\; \frac{r_i}{E^{(p)} - D_i}, \qquad D_i \approx \tilde{H}_{ii}. \tag{2}$$

The original code computed, for CSF column c of a block,

$$D_c^{\mathrm{old}} \;=\; \sum_r V_{rc}^2\, H_{rr}, \tag{3}$$

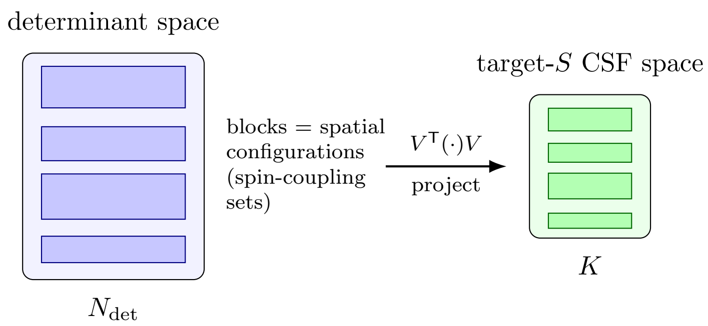

Figure 1: Option 2 in one picture. V is block diagonal by spatial configuration; each block is one spin-coupling set. The projection keeps only the target-S columns, so $K < N_{\text{det}}$ and every iterate is single-S by construction.

that is, the diagonal of $V^{\mathsf{T}} \operatorname{diag}(H)\, V$: it used only the *diagonal* entries of H. The exact quantity is

$$\boxed{D_c^{\text{new}} \;=\; \sum_r \sum_{r'} V_{rc}\, V_{r'c}\, H_{rr'}} \tag{4}$$

so (3) discards every $r \neq r'$ term. Those terms are not incidental. Within one block, r and r' label determinants of the *same* spatial configuration, differing only by a spin flip among the open shells — so $H_{rr'}$ is an exchange integral. These are large and predominantly negative, which means (3) is not a noisy estimate of (4) but a systematically biased one: $D^{\text{old}} > D^{\text{new}}$ for essentially every c.

Why the symptom looked like it did

Three observations that seemed puzzling all follow from one bias.

Spin stayed exact. V is exact and fixed, so every trial vector is a target-S combination no matter what D is. A bad preconditioner cannot contaminate the spin — it can only slow or stall the search. This is why the failure never presented as spin contamination, and why the usual purification remedies would not have helped.

Energies came out too high. With $D^{\text{old}} > D^{\text{new}}$ the denominators $E^{(p)} - D_i$ are wrong in a consistent direction, so the correction vectors are mis-scaled the same way at every iteration. The subspace stops making progress on a plateau, and Davidson reports the best Ritz value it has found. Since Ritz values are variational upper bounds, the result looks like a legitimately converged energy that is simply too high — there is no error flag.

It got worse with K. At small K the Krylov subspace grows to a sizeable fraction of the whole space and reaches the true root regardless of preconditioning. As K grows that brute-force route disappears and the preconditioner quality becomes decisive.

This is precisely the "spin-averaged preconditioner" requirement of Fales, Hohenstein and Levine (*J. Chem. Theory Comput.* **13**, 4162 (2017), Sec. 2.6): a determinantal CI preconditioner must average the exchange integrals *within a spin-coupling set.* A `ConfigBlock` in `single_spin.h` *is* one such set, and (3) is exactly the failure they warn about.

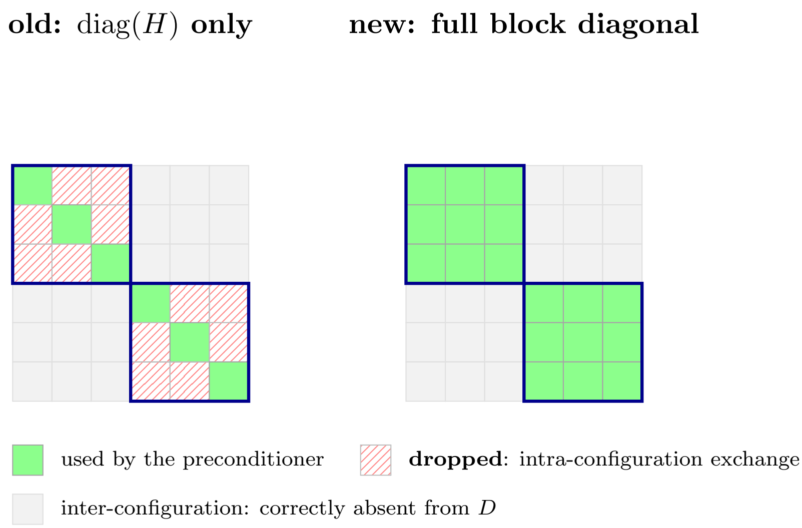

Figure 2: The determinant–determinant Hamiltonian, restricted to two spatial configurations (thick outlines). The old preconditioner used only the green diagonal cells; the hatched off-diagonal cells inside each block — the exchange elements between determinants of the same configuration — were dropped. They are large and mostly negative, so the omission biases every denominator in (2) the same way. Cells outside the blocks belong to different configurations and are legitimately not part of a *block*-diagonal preconditioner.

3 The fix

Exact block diagonal

`build_projected_diagonal()` evaluates (4) directly. For each block it forms the small dense determinant–determinant submatrix H_{bb} and contracts it with the block's CSF coefficients. Diagonal entries are taken from the precomputed `hii`; off-diagonal entries come from the general `Hij(DetA,DetB,...)` primitive in `chemistry/basic/determinants.h`, so SBD's own parity and phase conventions are reused rather than reimplemented.

Cost is $O(\texttt{max\_block\_dim} \times N_{\mathrm{det}})$ element evaluations, once, before the iteration. Block dimensions are $\binom{n_{\mathrm{open}}}{n_{\uparrow}}$ and small in practice; at $K = 113\,394$ the whole setup measured 5.2 s against a 52 s solve.

A blind alley, recorded so it is not retried. An earlier attempt batched this extraction into `max_block_dim` matrix–vector products, on the reasoning that blocks own disjoint determinant index sets, so one probe vector set to 1 on row r of *every* block would yield that row for all blocks at once. This is wrong: H couples determinants of *different* configurations, so each block's read-back is contaminated by cross-block elements. A numerical check showed errors of the same order as the matrix elements themselves. Direct `Hij` is both exact and cheaper.

Seeding

The seed was changed from the first n_{root} unit CSF vectors $e_0, \ldots, e_{n_{\mathrm{root}}-1}$ — an arbitrary corner of the space, being whichever configurations `std::map` happened to order first — to the n_{root} unit vectors with the *lowest* D_c. With the exact D these are the variationally best single-CSF starting guesses, and unit vectors are trivially orthonormal so nothing extra is needed.

A useful side effect: the Rayleigh quotient of a single unit CSF vector *is* its own projected diagonal, so iteration `0.0` now prints an energy equal to the reported seed `diag` value. That

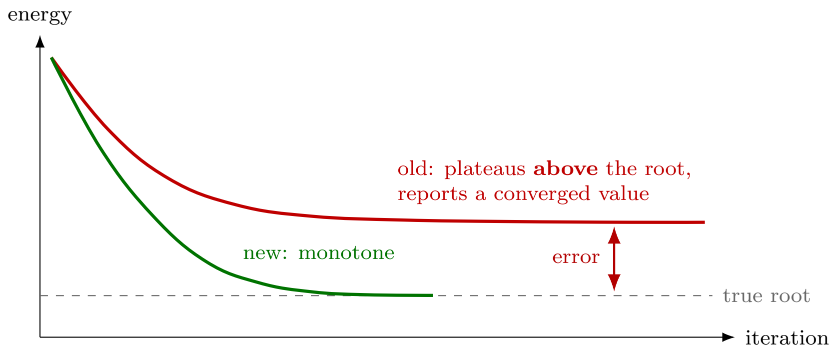

Figure 3: Schematic of the convergence behaviour (not measured data). The old preconditioner biases every correction the same way, so the iteration settles on a plateau above the true root and reports that value as converged; being a Ritz value it is a legitimate variational upper bound, so nothing signals an error. With the exact block diagonal the descent is monotone.

identity fails under (3), which makes it a free per-run check that the exact diagonal is in use.

Guard

A configuration whose open-shell count has the wrong parity for the requested S_z is now skipped rather than silently truncating $n_\uparrow = (n_{\text{open}} + 2S_z)/2$ by integer division.

4 Effect

Table 1: Sampled N_2, $R = 2.0$, 6-31G, 16 orbitals / 10 electrons, singlet, 3 roots, tolerance 10^{-8}, 4 threads, stored-matrix path. Energies are bit-identical before and after; only the iteration count changes.

	inner steps		
K	old	new	Davidson time (old → new)
56	31	21	0.0037 → 0.0028 s
117	53	28	0.0071 → 0.0042 s
3 906	251	39	1.055 → 0.196 s (5.4×)

The speedup grows with K, which is the signature of a preconditioner problem rather than a constant overhead. At $K = 113\,394$ — a case that previously stalled or returned nonsense — the solve now converges monotonically in **25 matrix–vector products** over 2 restart cycles.

5 Verification

- **Formula.** The direct-`Hij` construction was checked against a brute-force $V^\top HV$ diagonal in double precision: agreement to 2×10^{-16}.
- **Energies.** An independent dense diagonalisation of H and $\hat{S}^2$ over the same determinant list (built with the project's own Slater–Condon engine, outside SBD) reproduces the projected energies to all printed digits at $K = 56$, 117 and 3 906. At $K = 3\,906$ the exact lowest singlets are -108.792795252, -108.629192405, -108.588760240, matching the solver.

- **Cross-checks at scale.** At $K = 113\,394$ the Davidson energy, the explicit $\langle\psi|H|\psi\rangle$, and the trace of the one- plus two-body RDMs agree to $\sim 10^{-13}$. The RDM route is independent of the solver path, so it also validates the CSF $\to$ determinant rotation back.
- **Spin.** Converged roots are single-S to machine precision, as they were before — the fix does not touch spin purity, only the energy.

An interpretation trap worth flagging. In these sampled N_2 spaces at $K \leq 3\,906$ the variational ground state is a *quintet* ($\langle\hat{S}^2\rangle = 6$), not a singlet. Single-spin E_0 therefore correctly equals multi-spin E_1, and comparing it against multi-spin E_0 makes a correct result look broken. The ordering is size-dependent: at $K = 113\,394$ the singlet *is* lowest. Always compare against the lowest multi-spin root with matching $\langle\hat{S}^2\rangle$.

6 Where the code lives

`include/sbd/chemistry/gdb/single_spin.h`	
`build_projected_diagonal()`	exact D, Eq. (4)
`_davidson_projected_core()`	uses D; falls back with a warning
seeding block	lowest-D unit vectors
`build_config_projector()`	V, plus the S_z parity guard
`include/sbd/chemistry/basic/determinants.h`	
`Hij(DetA,DetB,...)`	element primitive reused for H_{bb}

The old approximation is retained as a fallback for callers that supply no exact diagonal; it prints a warning that energies may converge above the true root. Per-phase timings are available with `SBD_SS_TIMING=1`; note that the printed `end davidson` interval brackets the projector build and the per-root post-processing as well as the solve, so it is not the solve time when `nroots > 1` and `--rdm 1`.